Exp.No:07

24.9.26

Sliding window

Aim:

Write a program to implement flow control at data link layer using sliding window protocol.

Sender Program:

1. Input window size from the user
2. Input a text message from the user.
3. Consider 1 character per frame.
4. Create a frame with following fields
5. Send the frames & wait for acknowledgement from Receiver
6. Poder a file called Receiver-Buffer.
7. Check ACK field for the Acknowledgement number
8. If the Acknowledgement number is as expected, send new set of frames accordingly,

Else if NACK is received,

resend the frames accordingly.

Receiver File :

1. Reader a file called Sender-Buffer
2. Check the Frame no
3. If the Fame no are as expected, write the appropriate ACK no. in the Receiver-Buffer file. Else with NACK no. in the Receiver Buffer file.

Code :

Sender Program :

```
import time
sender = "Sender-Buffer.txt"
receiver = "Receiver-Buffer.txt"
w = int(input("Enter window size."))
msg = input ("Enter Message:")
frames = [(i,ch) for i, ch in
enumerate(msg)]
i = 0
while i < len(frames):
    with open(sender, "w") as f:
        for no, data in frames[i:i+w]
            f.write(f"{no}, {data}\n")
    print("Sending :", no, data)
```

```
time.sleep(2)
with open(receiver, "r") as f:
    type, num = f.read().split(",")
num = int(num)
if typ == "ACK":
    print("ACK received", num)
    i = num + 1
else:
    print("NACK received:", num)
    i = num.
print("Transmission completed")
```

Receiver Program:

```
import time
sender = "Sender_Buffer.txt"
receiver = "Receiver_Buffer.txt"
expected = 0
while True:
    try:
        with open(sender, "r") as f:
            frames = f.readlines()
    except:
        continue
    for line in frames:
        no, data = line.strip().
                   split(",", 1)
        No = int(no)
```

```
print ("Received :", no, data)
if no == expected:
    expected += 1
else:
    print ("Wrong Frame!")
    with open (receiver, "w") as f:
        f.write (f"NACK, {expected}")
    break
else:
    with open (receiver, "w") as f:
        f.write (f"ACK, {expected-1}")
time.sleep(2)
if expected >= 1000:
    break
```

Input:

Window size : 3

Message : HELLO

Output:

```
Sender:
Sending : 0 H          Sending : 3 I
Sending : 1 E          Sending : 4 0
Sending : 2 L          ACK received : 4
ACK received : 2
                       Transmission Completed
```

Result:

Thus, Sliding window is implemented successfully.